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Promotions of online reviews from a channel perspective

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ABSTRACT

The impact of promoting online reviews on pricing strategies and profits of a supply chain with one manufacturer and one retailer is modeled: channel members can improve their average rating in a reasonable range by promotional strategies with some costs. In the promotion process, a retailer can either increase or decrease the variance. Two channel structures, centralized and decentralized, are examined. By comparing the results with and without review promotion, it is discovered that a change of variance may enhance or undermine the effect of review promotion. First, a reduced variance can always increase the demand of the product but may hurt the profits of the manufacturer and retailer, regardless of the channel structures. Second, with an increased variance, both the manufacturer and the retailer gain more profits, but the efficiency of the promotion may decrease with the quality level. Third, under the decentralized structure, the manufacturer has a greater threshold interval to benefit from review promotion than the retailer, but the promotion strategy may not always play a positive role in improving the channel efficiency of the decentralized channel.

1. Introduction

It is widely acknowledged that consumers today rely heavily on online reviews to make purchasing decisions. According to Anderson (2014), 88% of consumers read online reviews to evaluate the quality of a local business. Recognizing the widespread influence of online reviews, firms adjust their marketing strategy in response. Given the fact that the online environment is anonymous and the informativeness of online reviews may be affected by some self-interested factors or external factors (Chaiken and Maheswaran, 1994; Ireland and Pennebaker, 2010), more firms are paying attention to the promotion of online reviews, in different forms. For instance, many retailers on Taobao.com or Jingdong.com offer financial incentives, rewards or coupons to consumers to encourage online reviews. This phenomenon is inevitably receiving increasing attention from academic researchers. Burtch et al. (2018) show that a combination of financial incentives and social norms can stimulate online reviews. Yang and Dong (2018) focus on the retailer's rebate strategy and study its impact on the pricing decision of the retailer. However, these and most related studies are empirical and only focus on a single-vendor scenario.

In practice, many products are delivered through a distribution channel with manufacturers and retailers. When a retailer on Taobao.com uses coupons or cash rewards to boost positive reviews, these promotion strategies inevitably affect consumers' valuation of the product (Krishnan et al., 2004), which further influences the demand for the product, and different demand patterns then bring about different supply chain dynamics (Huang et al., 2021). Thus, understanding the role of promotions is not only important for retailers but also for manufacturers.

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A supply chain with one manufacturer and one retailer is considered; the manufacturer distributes a single product to the retailer and the retailer sells the product to end consumers. Two channel structures, centralized and decentralized, are analyzed. In a centralized system, the manufacturer and retailer work as one to maximize the total profit. In a decentralized system, the manufacturer and retailer make their own decisions to maximize their individual profits. Note that we focus on experience products, whose qualities are not easily verifiable before their use (Nelson, 1974), such as books or shoes. Various studies have shown that for such products, consumers' purchasing decisions are significantly affected by online reviews (Zhao et al., 2013; Jiang and Yang, 2019). In particular, two metrics of online reviews, average rating (review valence) and review variance, are considered. It is generally agreed that average rating measures consumers' average assessment of the value of the product, which can significantly reflect the "quality level" of the product (Fainmesser et al., 2021; Li, 2017). For instance, a product with a higher quality is more likely to receive a higher average rating (e.g., 4 out of 5) while a product with lower quality may receive a lower average rating (e.g., 2 out of 5). Therefore, it is more practical for firms to implement promotion strategies to improve consumers' satisfaction and thus increase the average rating to a reasonable degree (Dellarocas, 2006).

It is worth noting that, following Feng et al. (2019), we consider more legal means of promoting online reviews. That is, the retailer can take some reasonable strategies to increase the value of the product to consumers at a cost, which then has a positive effect on the average rating. That is, they can either respond to those consumers who give unfavorable reviews or those who give favorable reviews. Take a digital camera with an average rating of 3 out of 5 (e.g., 1, 2, 5, 3, 4) as an example. The consumer who gives 1 star is highly unsatisfied with the product, so the retailer can strategically respond to that consumer and try to increase his/her satisfaction (Gu and Ye, 2014; Yang and Dong, 2018). For instance, it is common to find those follow-up reviews on the websites of Taobao.com. That is, consumers can choose to update their reviews in the review system. Consumers are likely to give a higher rating in their follow-up reviews when the retailer give them some satisfactory responses, such as some rebates or gifts.[1] In addition, some consumers may give the negative comments to the retailer to ask for some satisfactory feedback before posting the ratings.

In contrast, the consumer who gives 4 stars may only have a small unfavorable concern about the product and some rebates or added services may be helpful to increase his/her satisfaction. It is easy to find that the retailer can raise 1 to 2, which decreases the variance, or raise 4 to 5, which increases the variance. Therefore, based on this consideration, we explore the following questions: (1) Does a change of variance affect the optimal promotion strategy of the channel? (2) Is it always wise for the channel with different "quality levels" to promote the average rating? (3) Under different channel structures, can the manufacturer and the retailer always gain more profit from the promotion of online reviews?

The following findings will be later presented in detail. First, a decreased variance has a positive effect in increasing the demand for the product but may have a negative effect on the profits of the manufacturer and the retailer, regardless of the channel structure. In particular, a channel distributing a lower-quality product is more likely to be hurt by a promotion if the promotion cost is not small. Second, an increased variance may hurt the demand for the product in the promotion process and the demand in the centralized channel is more easily to be hurt by review promotions. This finding is different from that of Yang and Dong (2018), who consider the single-vendor scenario (i.e., the retailer) and indicate that review stimulation increases the demand. Interestingly, the players can always gain more profit with an increased average rating and variance, although the promotion efficiency may be decreased if the promotion cost is relatively high, especially for a channel with a higher-quality product. Third, we find that an increased variance can even improve the channel efficiency of a decentralized channel, but a decreased variance may lower the channel efficiency.

The rest of this paper is organized as follows. Section 2 reviews the prior literature. In Section 3, the model is presented. Discussion of the implications of promotions under the centralized and decentralized structures is given in Sections 4 and 5. Section 6 compares the results for the two channel structures and Section 7 presents numerical studies that verify the propositions.Section 8 conducts several model extensions. Conclusions and future work are presented in Section 9.

2. Literature review

This study is mainly aligned with three streams in the literature: online reviews, review promotion, and pricing and effort decisions in supply chain management. In the literature on online reviews, a wealth of publications empirically study the relationship between product sales by focusing on different metrics, such as the average rating (review valence), review variance, and review volume. In particular, the average rating and variance show the distribution of reviews and thus serve as cues to marketing decisions (Langan et al., 2017). A number of studies support the idea that the high valence can increase the intent of consumers to purchase (Chevalier and Mayzlin, 2006; Chintagunta et al., 2010). However, there have been inconsistent findings in terms of the variance. For instance, Babić Rosario et al. (2016) find that the variance of the reviews has a negative effect on consumer purchase intention, while Clemons et al. (2006) find that the dispersion of ratings has a positive relation with sales. Sun (2012) studies the informational role of ratings and concludes that the average rating indicates the quality of the product, whereas the variance reflects the type of the product (niche product or mass product). Apart from examining the impact of online reviews on sales, the effect of online reviews on firms' strategies has been much discussed. Chen and Xie (2005, 2008) treat online reviews as marketing communications and claim that firms should adjust their marketing strategies in response to online reviews. Li et al. (2011) focus

[1] https://detail.tmall.com/item.htm?spm=a230r.1.14.61.7b4521e1ZmFptI&id=624314022460&ns=1&abbucket=10.

on repeat purchases of products and examine the influence of online reviews on the profitability of the firm. Liu et al. (2017) study the joint impact of online reviews and sales volume on a firm's pricing strategy.

In the literature on promotion or manipulation of reviews, numerous studies have shown that this is a growing practice in different areas, such as books (Northrup, 2009), music (Mayzlin, 2006) and tourism (Gössling et al., 2018). It has also been explored how review manipulation affects the consumers' purchase decisions and the firms' strategies. Dellarocas (2006) points out that strategic manipulation of online reviews can increase the information value of the reviews to consumers under some conditions. Mayzlin et al. (2014) empirically examine the effect of promotional reviews by comparing the reviews on Expedia.com and TripAdvisor.com. Yang and Dong (2018) study rebates for stimulating online reviews and suggest that a rebate strategy can expand demand. Burtch et al. (2018) show that offering financial incentives can stimulate consumers to give more favorable reviews. They point out that businesses with low reputations are more likely to manipulate reviews.

It is notable that most of the studies mentioned above are empirical and focus on a single-vendor scenario. In practice, numerous products are delivered through a distribution channel with manufacturers and retailers. Therefore, the interaction between the manufacturer and the retailer plays a significant role in supply chain management (Choi et al., 2020; Yang et al., 2020; Li et al., 2019, 2020). We have been inspired by the literature considering the marketing efforts of the members of a channel in a supply chain. Some work focuses on the efforts of these members to affect the demand through different strategies, such as advertising and services (Gou et al., 2020; Karray, 2015; Li et al., 2016). In these studies, coordinating contracts for sharing costs or revenues are important in supply chain management. Krishnan et al. (2004) study the coordination of contracts for a supply chain with one manufacturer and one retailer, in which the retailer decides on the promotional effort. Kunter (2012) focuses on the coordination in a manufacturer–retailer channel via cost and revenue sharing contracts. In addition, cooperative advertising is treated as a powerful strategy in marketing channels in which one party undertakes a certain fraction of the advertising expenditure for its partner. This practice has been widely addressed in the literature (Ahmadi-Javid and Hoseinpour, 2018; Gou et al., 2020).

The present paper extends the above literature by focusing on review promotions from a channel perspective. First, in line with the model framework of Li (2017) and Sun (2012), we focus on the distribution of reviews and theorize that consumers' value is affected by the variance and average rating. Second, we contribute to the literature on review promotions. More specifically, our results show that the promotion strategy may hurt the demand of the supply chain, which is different from the findings of Yang and Dong (2018), who suggest that review stimulation increases the demand. Third, although some studies consider the impact of online reviews from a channel perspective, there is little about the promotion of online reviews from a channel perspective. To fill this gap, we discuss the promotion of online reviews in a distribution channel consisting of one manufacturer and one retailer. We investigate whether such promotion is always wise for the retailer and whether the manufacturer can be better off with this strategy.

3. The model

Consider a manufacturer–retailer supply chain in which the manufacturer sells an experience product to the retailer at a unit wholesale price w and the retailer sells the product to end consumers at a unit retail price p. For this product, the quality is not readily perceived by consumers. Thus, in line with Jiang and Yang (2019), we assume that consumers rely on online reviews to make their decisions. Further, we assume that the product has two dimensions: a vertical dimension (e.g., quality) and a horizontal dimension (e.g., color or size). Specifically, the quality captures the attribute of the product whose valuation is the same for all consumers, and a higher quality always means a higher willingness-to-pay (Dellarocas, 2006). For example, consumers all prefer digital cameras with a better battery durability. A horizontal component reveals inconsistencies in the valuations by consumers. To illustrate, consumers may want different colors when they purchase smart phones or clothes. We use q_i to indicate the perceived value of the product to consumer i, which is affected by the vertical dimension and the horizontal dimension. Thus, the utility of the product to consumer i is $q_i - p$, where p is the price of the product. Consistently with Sun (2012) and Li (2017), we further assume that q_i follows a uniform distribution $[q - a, q + a]$, where q represents consumers' average valuation towards the product while the variance a indicates consumers' inconsistency in their valuation of the product. Thus, consumer i would buy the product if $q_i > p$. Then, the demand function can be characterized as follows:

$$D = \frac{q + a - p}{2a}. \tag{1}$$

Next, the case of review promotions is derived. Following Dellarocas (2006), we assume that the retailer can encourage consumers to increase the average rating of the product, at some cost. In this process, the retailer can choose to increase or decrease the variance (from a to b) by targeting different consumer groups. Here, we assume that b is an exogenous variable, which can be higher or lower than a. For instance, for a product with the ratings 1, 1, 5, 4, 3, 4, it is possible for the retailer to affect the consumers who give relatively unfavorable reviews (from 1 to 2) or who give relatively favorable reviews (from 4 to 5), which corresponds to a lower variance or a higher variance. In practice, in the current review system, consumers can choose to give the follow-up reviews after the first review, and thus it is reasonable for the retailer to response to some consumers to affect their follow-up ratings. In the long term, these reviews can be treated as a steady signal to indicate the evaluations of consumers. Following Kwark et al. (2014), we do not consider this dynamic process and instead examine the impact of promotions when online reviews are in a steady state. Note that we restrict the range of review promotions to ensure that the retailer only affects a small range of consumers, so that we can focus on the effect of the quality spectrum on the efficiency of the promotion. More specifically, the retailer can increase consumers' perception of the value of the product by increasing the average rating of the online reviews from q to $q + \eta$ at a cost of $\frac{c}{2}\eta^2$, where c captures the cost efficiency of promotions of online reviews, and η reflects the increased value brought by the promotion strategy, such as some rebates, coupons or free after-sales services.

Table 1
Model specification of a centralized channel.

	No review promotion	With review promotion
Decision variables	p_c	p_c^M, η
Demand functions	$D_c = \frac{q+a-p_c}{2a}$	$D_c^M = \frac{q+\eta+b-p_c^M}{2b}$
Channel profits	$\pi_c = D_c p_c$	$\pi_c^M = D_c^M p_c^M - \frac{c\eta_c^2}{2}$

Therefore, with the promotion strategy, for consumer i, the perceived value of the product follows the uniform distribution $[q+\eta-b, q+\eta+b]$. Then, the demand function with promotions of online reviews can be obtained as

$$D^M = \frac{q+\eta+b}{2b} - \frac{p}{2b}. \tag{2}$$

We normalize the size of the market to 1 and assume that the market is never fully covered, either with or without review promotions. As mentioned by Sun (2012), sufficient conditions for incomplete coverage of the market both with and without review promotions are $\frac{q+a}{2a} < 1$ and $\frac{q+\eta+b}{2b} < 1$ (see Eqs. (1) and (2)). Thus, for convenience, we have $0 < q \le 1, 0 < \eta \le 1, a > 2$ and $b > 2$ throughout this paper. Here, we follow Li (2017) and allow for the possibility that consumers' valuations of the product can be negative (i.e., $q < a$ and $q + \eta < b$). For example, consumers may be very unsatisfied with a book with violent content or may be hurt by uncomfortable shoes. Moreover, this assumption has also been verified by Wu et al. (2021).

In the following, based on these two demand functions, we further analyze the impact of review promotions on the pricing and profits of the manufacturer and the retailer, under different structures.

4. Promotions of online reviews under a centralized structure

In this section, we consider the channel integration, that is, the manufacturer and the retailer act as one to maximize the total profit of the channel. To better understand the impact of review promotions, we treat the case without promotions of reviews as the benchmark. In this scenario, the manufacturer and the retailer set the retail price (p_c) to maximize the channel profit (π_c). For simplicity, we assume that the marginal cost for each demand is zero. Then, in the presence of promotions, the manufacturer and the retailer decide the retail price (p_c^M) and the degree of review promotions (η_c) at the same time to maximize the channel profit π_c^M, at a cost of $\frac{c}{2}\eta_c^2$. Specifically, with the above demand functions (Eqs. (1) and (2)), we can formulate the profit without review promotions (π_c) and with review promotions (π_c^M). Table 1 gives a clear model specification for the centralized structure.

We first give the equilibrium results with and without review promotions in the centralized structure in the following lemmas.

Lemma 1. *The equilibrium retail price, the demand, and the channel profit without promotions of online reviews in the centralized structure are as follows.*

$$p_c = \frac{q+a}{2}, \quad D_c = \frac{q+a}{4a}, \quad \pi_c = \frac{(q+a)^2}{8a}.$$

Lemma 2. *The equilibrium retail price, the demand, and the channel profit with promotions of online reviews in the centralized structure are as follows.*

$$\eta_c = \frac{q+b}{4bc-1}, \quad p_c^M = \frac{2(q+b)bc}{4bc-1}, \quad D_c^M = \frac{(q+b)c}{4bc-1}, \quad \pi_c^M = \frac{(q+b)^2 c}{2(4bc-1)}.$$

The proofs are given in the Appendix A.

Note that we aim to analyze the relationship between η_c and q and study the efficiency of the promotion strategy from the perspective of product quality, we add the following constraints throughout this paper: $c \ge \frac{2+b}{4b}$. In fact, it is easy to verify that the profit function is always increasing in η_c if $c < \frac{2+b}{4b}$, which is not realistic. Therefore, this condition helps us to avoid such trivial cases and ensures that $\eta_c \le 1$ always holds, which guarantees that optimal solutions are obtainable and keeps the analysis simple without affecting the key findings. A similar assumption can be found in the model of Liu et al. (2018). With the same logic, Martín-Herrán and Sigué (2017) take $c = 1$ in their models. Moreover, we introduce $X_1 = \frac{c}{4bc-1}$ in the rest of this paper for notational simplicity. By comparing the equilibrium results in Lemmas 1 and 2, we derive the implication of review promotions under the centralized structure.

Proposition 1. *Under the centralized structure,*

(1) The optimal degree of promotion (η_c) increases with product quality (q) and decreases with b.
(2) When $b > a$, the retail price with review promotions is always higher (i.e., $p_c^M > p_c$). When $b < a$, the retail price with review promotions is lower if and only if $c > \frac{a}{4ab-4b^2}$ and $q < min\{\frac{a-4b^2X_1}{4bX_1-1}, 1\}$.
(3) When $b < a$, the demand with review promotions is always higher. When $b > a$, the demand with review promotions is lower if and only if $c > \frac{a+1}{4(b-a)}$ and $\frac{a(4bX_1-1)}{1-4aX_1} < q < 1$.

Table 2
Model specification of a decentralized channel.

	No review promotion	With review promotion
Manufacturer's decisions	w	w^M, λ
Retailer's decisions	p_d	p_d^M, η
Demand functions	$D_d = \frac{q+a-p_d}{2a}$	$D_d^M = \frac{q+\eta+b-p_d^M}{2b}$
Manufacturer's profits	$\pi_m = D_d w$	$\pi_m^M = D_d^M w^M - \frac{c}{2}\lambda\eta_d^2$
Retailer's profits	$\pi_r = D_d(p_d - w)$	$\pi_r^M = D_d^M(p_d^M - w^M) - \frac{c}{2}(1-\lambda)\eta_d^2$

The proof is given in the Appendix A.

Proposition 1 first shows that the optimal degree of promotion is a linear function of the quality of the product, and the higher the quality, the higher the optimal degree of promotion. This result is consistent with the findings of Dellarocas (2006). The reason is that the marginal profit of the supply chain is a monotonically increasing function of the true quality, and thus the retailer with higher quality product may have a higher incentive to promote the reviews. However, the optimal degree of promotion decreases with the variance. This result is also in line with the result of Sun (2012), who calculate the average rating (M) and the variance (V) as $M = \upsilon - \sqrt{3V}$. A possible intuition for this is that a higher variance actually shows that there is a greater diversity of opinion among the consumers, which may negatively affect their perception of the value of the product. Thus, if the channel members can increase the variance to a large extend, it is better to improve the average rating slightly.

Proposition 1 then demonstrates that review promotions may not always increase the retail price. Specifically, if review promotions yield an increased variance, the retailer can charge a higher price. This is because, with a higher variance, consumers may become less sensitive to price changes, and the promotion strategy increases the consumers' valuation of the product, which drives the retailer to increase the price. However, if the variance increases, the promotion strategy may hurt the demand for the product if the product has a relatively high quality. In this case, the promotion strategy plays a role in showing an increased inconsistency in the consumers' valuations of the product. That is, consumers may face a higher uncertainty about whether the product matches their needs or preferences, and this reduces the demand for the product. This result is consistent with West and Broniarczyk (1998), who find that a higher variance increases the likelihood of purchase if and only if the average rating is below an aspirational level.

In contrast, if the variance is reduced, whether the retailer can set a higher price or a lower price depends on the promotion cost efficiency and the product quality. Specifically, if the quality of the product is quite low, the positive effective of a promotion strategy in increasing consumers' willingness to pay is not obvious, but consumers may be more sensitive to the change of price. Therefore, a retailer selling a low-quality product would have a higher incentive to cut the retail price to increase the demand.

Proposition 2. *Under the centralized structure,*

(1) When $b > a$, the whole supply chain is always better off with review promotions (i.e., $\pi_c^M \geq \pi_c$), but the promotion efficiency decreases with the product quality (i.e., $\frac{\partial(\pi_c^M - \pi_c)}{\partial q} < 0$) if and only if $c > \frac{a+1}{4(b-a)}$ and $q > \frac{a(4bX_1-1)}{1-4aX_1}$.

(2) When $b < a$, the whole supply chain is better off if $c < \frac{a}{4b(a-b)}$; otherwise, the whole supply chain is worse off with review promotions if $q < min\{q_c, 1\}$, where $q_c = \frac{a-2b\sqrt{aX_1}}{2\sqrt{aX_1}-1}$.

The proof is given in the Appendix A.

Proposition 2 indicates that the channel can gain more profits with an increased variance, regardless of the promotion cost or the quality level of the product. A possible reason could be as follows. With a higher variance, consumers becomes less sensitive to the product price, thus a moderate increase of the average rating drives the retailer to charge a higher price, which leads to an increase of the channel profit. However, the promotion efficiency may decrease if the promotion cost coefficient is relatively high. In this case, the high promotion cost undermines the efficiency of the promotion, especially for a product with a higher quality. This is because the retailer tends to implement a higher degree of review promotions, which makes the corresponding promotion cost also higher than for a retailer selling the product with a lower quality. In contrast, if the promotion leads to a lower variance, the supply chain may be worse off if the cost coefficient is relatively high ($c > \frac{a}{4b(a-b)}$). Besides, a channel distributing a product with a lower quality has a higher probability of being hurt than a channel distributing a product with a higher quality. As analyzed before, when the variance is reduced, the positive effect of increasing consumers' willingness to pay is too small to outweigh the promotion cost, especially for a low-quality product, which reduces the profit of the channel.

5. Promotions of online reviews under a decentralized structure

In this section, we consider a decentralized channel structure, that is, the manufacturer and the retailer aim to maximize their own profits. In the absence of promotions of online reviews, the sequence of the game is as follows. The manufacturer first decides the wholesale price w to maximize its profit π_m, and the retailer then determines the retail price p_d to maximize its profit π_r, given the wholesale price w. Thus, the manufacturer's profit π_m and the retailer's profit π_r without promotions of online reviews can be easily obtained. Table 2 also gives a clear model specification for this decentralized channel.

Then, in the presence of promotions of online reviews, we follow Lu et al. (2019) and assume that the manufacturer provides a subsidy proportion λ and the retailer decides the degree of the promotion strategy η_d. The sequence of the game is as follows. The manufacturer first decides the wholesale price w and the subsidy proportion λ to maximize its profit π_m^M, and then the retailer is presented as the follower to decide the retail price p_d and the degree of promotion η_d to maximize its profit π_r^M (See Table 2). We next give the equilibrium outcomes in the decentralized supply chain with and without review promotions in the following lemmas.

Lemma 3. *The equilibrium wholesale price, retail price, demand, and profits of the manufacturer and the retailer without review promotions in the decentralized structure are as follows.*

$$w = \frac{q+a}{2},\quad p_d = \frac{3(q+a)}{4},\quad D_d = \frac{q+a}{8a},\quad \pi_m = \frac{(q+a)^2}{16a},\quad \pi_r = \frac{(q+a)^2}{32a}.$$

Lemma 4. *The equilibrium wholesale price, retail price, demand, and profits of the manufacturer and the retailer with review promotions in the decentralized structure are as follows.*

$$\eta_d = \frac{6(q+b)}{32bc-9},\quad \lambda = \frac{1}{3},\quad w^M = \frac{(q+b)(16bc-3)}{32bc-9},\quad p_d^M = \frac{3(q+b)(8bc-1)}{32bc-9},$$
$$D_d^M = \frac{4c(q+b)}{32bc-9},\quad \pi_m^M = \frac{2c(q+b)^2}{32bc-9},\quad \pi_r^M = \frac{4c(8bc-3)(q+b)^2}{(32bc-9)^2}$$

The proofs are given in the Appendix A.

It is interesting to note that the sharing rates of the manufacturer and the retailer are constants and do not vary with the degree of the review promotions. The reason is the following. The manufacturer's marginal profit is independent of λ while the retailer's marginal profit is increasing in λ. Therefore, it is possible for the manufacturer to give a constant participation rate to ensure that the retailer is willing to take the promotion strategy. Moreover, the retailer tends to share more costs than the manufacturer if taking the promotion strategy.

Similar to the results under the centralized channel, we find that the optimal degree of promotion under the decentralized channel is increasing in q but decreasing in b. In addition, η_d is always smaller than η_c, which means that under the decentralized channel, the retailer may have less motivation to increase the degree of promotion. Taking the above results into account, the impact of promotion on the pricing decisions and performance of the manufacturer and retailer are examined next. For simplicity, we introduce $Y_1 = \frac{16bc-3}{32bc-9}, Y_2 = \frac{3(8bc-1)}{32bc-9}, Y_3 = \frac{2c}{32bc-9}, Y_4 = \frac{4c(8bc-3)}{(9-32bc)^2}$ in the rest of the paper.

Proposition 3. *Under the decentralized structure,*

(1) When $b \geq a$, the wholesale price with review promotions is always higher; When $b < a$, the wholesale price with review promotions is lower if and only if $c > \frac{3(3a-2b)}{32b(a-b)}$ and $q < min\{\frac{a-2bY_1}{2Y_1-1}, 1\}$.

(2) When $b \geq a$, the manufacturer is always better off with review promotions, but the promotion efficiency decreases with the product quality if and only if $c > \frac{9(a+1)}{32(b-a)}$ and $q \geq \frac{a(16bY_3-1)}{1-16aY_3}$; when $b < a$, the manufacturer is hurt if and only if $c > \frac{9a}{32b(a-b)}$ and $q < min\{q_m, 1\}$, where $q_m = \frac{a-4b\sqrt{aY_3}}{4\sqrt{aY_3}-1}$.

The proof is given in the Appendix A.

Proposition 3 first indicates that in the decentralized structure, the manufacturer may not always benefit from a promotion strategy. More specifically, when the variance decreases, the manufacturer who distributes a low-quality product is more likely to be hurt. As mentioned in the situation of the centralized structure, when the variance is reduced, the positive impact of the promotion on increasing consumers' willingness to pay is quite limited, especially for a low-quality product, which may drive the manufacturer to cut the wholesale price, leading to a reduced profit. Proposition 3 then demonstrates that when the variance increases, the efficiency of the promotion may decrease, especially for a high-quality product. The intuition is that an overly high variance may make consumers doubt the high quality of the product, which lower their willingness to pay and thus reduces the demand for the product to a large extent, negatively affecting the promotion efficiency.

Next, we derive the impact of review promotions on the pricing and profitability of the retailer, and give further explanations.

Proposition 4. *Under the decentralized structure,*

(1) When $b \geq a$, the retail price with review promotions is always higher; When $b < a$, the retail price with review promotions is lower if and only if $c > \frac{9a-4b}{32b(a-b)}$ and $q < min\{\frac{3a-4bY_2}{4Y_2-3}, 1\}$.

(2) When $b \geq a$, the retailer is always better off with review promotions, but the promotion efficiency decreases with the product quality if and only if $c > \frac{3}{32}\left(\frac{a+1}{b-a} + \frac{2}{b} + \frac{\sqrt{a(b+1)(a(b+4)-3b)}}{b(b-a)}\right)$ and $q \geq \frac{a(32bY_4-1)}{1-32aY_4}$; When $b < a$, the retailer is worse off if and only if $c > \frac{3}{32(a-b)}\left(\frac{3a-2b}{b} + \sqrt{\frac{4b-3a}{b}}\right)$ and $q < min\{q_r, 1\}$, where $q_r = \frac{a-4b\sqrt{2aY_4}}{4\sqrt{2aY_4}-1}$.

The proof is given in the Appendix A.

Proposition 4 implies that the retailer can benefit from the promotion if the variance increases, but an overly high variance may reduce the efficiency of review promotions, especially for a product with a relatively high quality. The reason is quite similar to that

under the centralized structure. Combined with Proposition 3, it is found that when the variance decreases, the promotion strategy may hurt both players' profits, especially for channel distributors with a relatively low-quality product. To further analyze who has a higher probability of gaining from the promotion strategy, we compare the threshold values in Proposition 4.

Corollary 1. *Under the decentralized structure, when $b < a$, the manufacturer gains but the retailer loses if and only if $\frac{9(a+1)}{32(b-a)} < c < \frac{9(a+1)^2}{32(a-b)(ab-1)}$ and $q_m < q < min\{q_r, 1\}$.*

Corollary 1 indicates that it is wise for the manufacturer to share the promotion cost if the retailer decides to take the promotion strategy. This is because the manufacturer has a greater threshold interval to benefit from the manipulation than does the retailer. This is because when $b < a$ and $\frac{9(a+1)}{32(b-a)} < c < \frac{9(a+1)^2}{32(a-b)(ab-1)}$, the marginal profit of the manufacturer is always higher but the marginal profit of the retailer may be hurt by the promotion strategy.

6. Comparisons between the two channel structures

In this section, we first compare the implication of the promotion strategy for the demand for the product under the two channel structures.

Proposition 5. *When $b < a$, the demand is always higher with review promotions, regardless of channel structure. When $b > a$, the demand under the centralized structure is hurt by review promotion if and only if $c > \frac{a+1}{4(b-a)}$ and $q > \frac{a}{4(b-a)c-1}$, while the demand under the decentralized structure is hurt by review promotion if and only if $c > \frac{9(a+1)}{32(b-a)}$ and $q > \frac{9a}{32(b-a)c-9}$.*

The proof is given in the Appendix A.

Proposition 5 states that when promotions of online reviews lead to an increased variance, this strategy exerts a negative effect on the product demand, regardless of the channel structure. In particular, when $c > \frac{a+1}{4(b-a)}$ and $\frac{a}{4(b-a)c-1} < q < \frac{9a}{32(b-a)c-9}$, the demand under the decentralized structure is increased but the demand under the centralized structure is decreased. In other words, the demand under the centralized channel has a higher probability of being hurt by the promotion. This is because the equilibrium degree of promotion (i.e., $\eta_c = \frac{q+b}{4bc-1}$) under the centralized structure is always higher than the equilibrium degree of promotion (i.e., $\eta_d = \frac{6(q+b)}{32bc-9}$). Thus, under the centralized channel, with review promotion, the channel has a higher probability of raising the retail price, which would decrease the demand.

Next, we are more interested in the impact of the promotion strategy on the performance of the decentralized channel. In particular, we use the ratio of the channel profit under the decentralized structure to the channel profit under the centralized channel to denote the channel efficiency of the decentralized structure. That is, the channel efficiency without the promotion and with the promotion can be characterized as $(\pi_m + \pi_r)/\pi_c$ and $(\pi_m^M + \pi_r^M)/\pi_c$, respectively. Then we have the following proposition.

Proposition 6. *When $b > a$, the promotion strategy can always improve the channel efficiency of the decentralized structure. When $b < a$, the promotion strategy hurts the channel efficiency if and only if $c > \frac{\sqrt{b(25b-9a)}+9a-5b}{32b(a-b)}$ and $q < min\{\frac{a(81-256bc)+8(a-b)(32bc-9)\sqrt{ac(16bc-5)}}{64ac(16bc-5)-(32bc-9)^2}, 1\}$.*

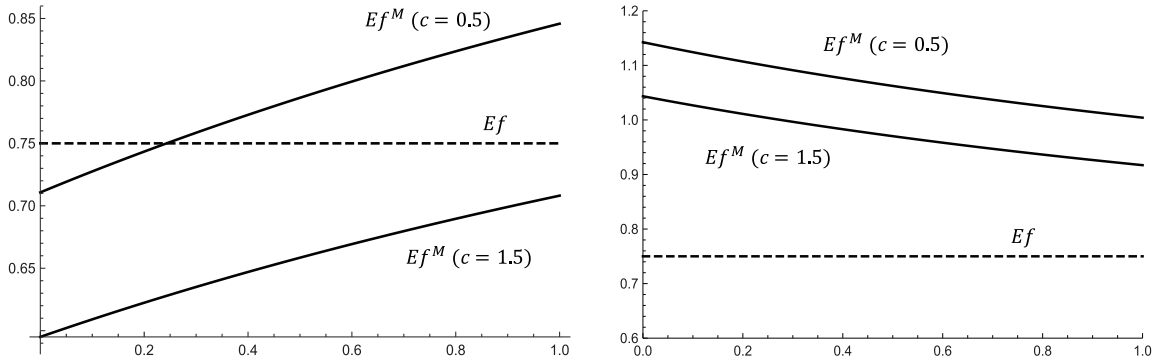
The proof is given in the Appendix A.

Fig. 1 illustrates the results of Proposition 6. Note that Ef^M and Ef denote the channel efficiency of the decentralized structure with and without the promotion, where $Ef = (\pi_m + \pi_r)/\pi_c$ and $Ef^M = (\pi_m^M + \pi_r^M)/\pi_c$. Proposition 6 and Fig. 1 show that the promotion strategy can improve the performance of the decentralized structure as long as the retailer increases the variance. This is because under this situation, it is better for the manufacturer and the retailer to increase the wholesale price and retail price. Although this may drive away marginal consumers, both players will still be better off because of the increased price. However, if the variance decreases, the promotion may negatively affect the channel efficiency. Therefore, it is not wise for the players under the decentralized channel to promote the average rating if the promotion cost is relatively high, especially for a channel distributing a relatively low-quality product.

7. Numerical studies

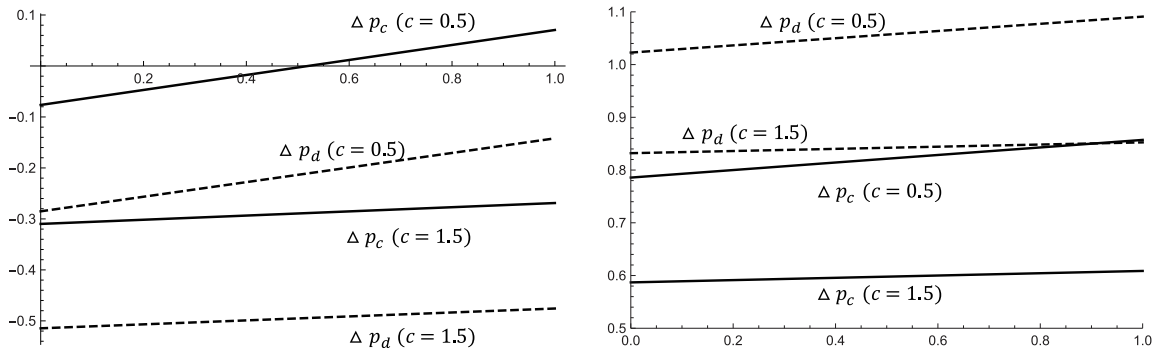
Numerical studies are conducted to analyze the impact of review promotions on the decision variables and profits of the members of the supply chain. In the following, we take $a = 3$. Fig. 2 shows the impact of b and q on the price difference of the two channel structures. Note that $\Delta p_c = p_c^M - p_c$ and $\Delta p_d = p_d^M - p_d$. It can be seen that if the promotion strategy reduces the variance, the channel selling a product with low quality should lower the selling price and the price under the decentralized channel then is lowered more. If the variance only changes slightly, we find that the retailer can raise the retail price under both channel structures, and the degree of increase is also similar. In addition, if the variance increases, the price under the decentralized channel increases more than the price under the centralized structure. In general, it is concluded that the sale price under the decentralized channel is more easily affected by review promotions.

Fig. 3 compares the impacts of b and q on channel profits, under the two channel structures, in which $\Delta\pi_c = \pi_c^M - \pi_c$ and $\Delta\pi_d = \pi_d^M - \pi_d$. It can be seen that a reduced variance may hurt the profit of the channel, regardless of the channel structure. Besides, a channel distributing a relatively low-quality product is more likely to be worse off in the presence of review promotions. However,



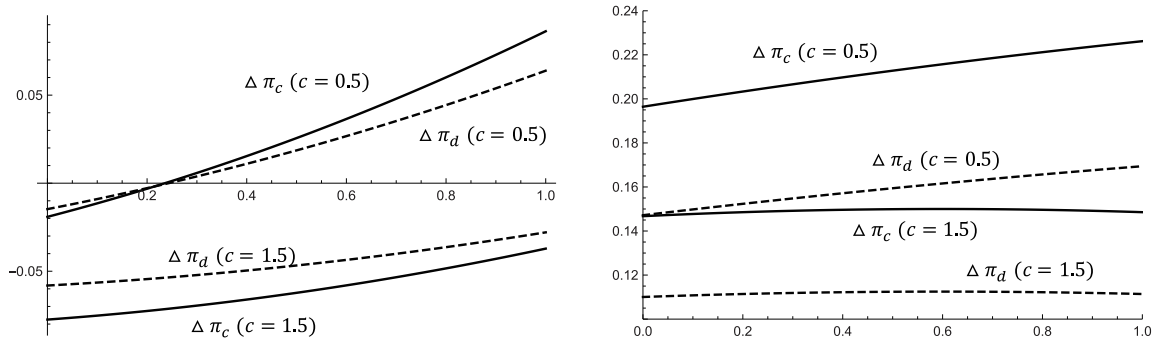
(1) $b = 2.2\ (b < a)$ (2) $b = 4\ (b > a)$

Fig. 1. The impact of review promotions on channel efficiency ($a = 3$).



(1) $b = 2.2\ (b < a)$ (2) $b = 4\ (b > a)$

Fig. 2. Impacts of review promotions on retail prices of players under the two channel structures.



(1) $b = 2.2\ (b < a)$ (2) $b = 4\ (b > a)$

Fig. 3. Impacts of review promotions on profits of players under the two channel structures.

under this case, the profit under the centralized channel decreases more than that under the decentralized channel, despite the fact that the retail price under the decentralized channel exhibits a greater decrease. In addition, as mentioned above, an increased variance always has a positive effect on the profits of the channel. However, if the promotion cost coefficient is relatively high, the promotion efficiency may decrease with the quality level. In other words, for a channel selling a product with excellent quality, the impact of review promotions on increasing the channel profit is quite limited.

8. Model extensions

In this section, in order to examine the robustness of our results, we extend the baseline model in two directions by considering (1) traditional consumers who are not influenced by online reviews and (2) smart consumers who are not fully influenced by the promotion. We demonstrate that the main insights are still hold under these model extensions.

8.1. Traditional consumers

We consider two types of consumers in the market. The first type of consumers, as discussed in the main model, relies on online reviews to make purchase decisions. The second type of consumers is not influenced by online reviews. For simplicity, we call the second type of consumers "traditional consumers" and assume that the perceived value q_s of traditional consumers is uniformly distributed over [0,1]. Thus, the consumer would buy the product if $q_s - p > 0$. Further, we assume that the size of second type of consumers is k and the size of first type of consumers is $1-k$. For simplicity, we only present the results of the centralized case, and the decentralized case has a similar result. The main results are summarized in Proposition 7. In addition, the proofs and the expression of q_e are presented in Appendix B.

Proposition 7. *Under the centralized structure,*

(1) When $b > a$, the retail price with review promotions is always higher (i.e., $p_c^M > p_c$). When $b < a$, the retail price with review promotions is lower if and only if $c > \frac{a-ak}{4ab-4b^2}$ and $q < min\{\frac{(k+1)(a(4bc+k-1)-4b^2c)}{k(8bc(a-b)+k-2)+1}, 1\}$.

(2) When $b > a$, the whole supply chain is always better off if and only if $0 < k < \sqrt{\frac{4a^2b^2+a^2-2ab+b^2}{(3ab-2a-2b+1)^2}} + \frac{a(-b)-a-b+1}{3ab-2a-2b+1}$, otherwise, the channel may be worse off if $c > \frac{(1-k)((a-1)k+a+1)^2}{4(b-a)(k^2(a(3b-2)-2b+1)+2k(ab+a+b-1)-ab+1)}$.

(3) When $b < a$, the whole supply chain is better off if $c < \frac{a-ak}{4ab-4b^2}$; otherwise, the whole supply chain may be worse off with review promotions if and only if $q < min\{q_e, 1\}$.

Proposition 7 first demonstrates that the pricing decision is still hold when we consider the traditional consumers who are not influenced online reviews. In addition, similar to the results in the basic model, a reduced variance may hurt the profit of the channel if the promotion cost is relatively high. However, the second result differs from the result in main model, suggesting that an increased variance helps the channel to gain more profit only when the proportion of traditional consumers in the market is not very large. Therefore, the channel having a considerable number of consumers who are not caring about online reviews should be more careful to take the promotion strategy.

8.2. Smart consumers

In the basic model, consumers are assumed cannot perceive the promotion of reviews. We now examine the case when consumers are smart to adjust the interpretation of online reviews in the presence of review promotions. In specific, when the promotion degree is η, the perceived value of the product will follow the uniform distribution $[(1-\gamma)q + \gamma(q+\eta) - b, (1-\gamma)q + \gamma(q+\eta) + b]$, where γ reflects the degree of consumers' myopia or naïveté. In other words, $(1-\gamma)$ measures the degree of consumers' smartness. Here, we only present the results under the centralized case, and decentralized case has a similar results. The main results are summarized in Proposition 8 and the proofs are presented in Appendix B.

Proposition 8. *Under the centralized structure,*

(1) When $b > a$, the retail price with review promotions is always higher (i.e., $p_c^M > p_c$). When $b < a$, the retail price with review promotions is lower if and only if $c > \frac{a\gamma^2}{4ab-4b^2}$ and $q < min\{\frac{4abc-a\gamma^2-4b^2c}{\gamma^2}, 1\}$.

(2) When $b > a$, the whole supply chain is always better off with review promotions (i.e., $\pi_c^M \geq \pi_c$), but the promotion efficiency decreases with the product quality (i.e., $\frac{\partial(\pi_c^M - \pi_c)}{\partial q} < 0$) if and only if $c > \frac{(a+1)\gamma^2}{4(b-a)}$ and $\frac{a\gamma^2}{4bc-4ac-\gamma^2} < q < 1$.

(3) When $b < a$, the whole supply chain is always worse off if $\gamma > \frac{1}{a+1}\sqrt{\frac{(b+2)(a-b)(ab-1)}{b}}$ or if $c > \frac{(a+1)^2\gamma^2}{4(a-b)(ab-1)}$; otherwise, the whole supply chain is better off if and only if $q > max\{0, \frac{2(a-b)\sqrt{ac(4bc-\gamma^2)}}{4ac-4bc+\gamma^2} - \frac{a\gamma^2}{4ac-4bc+\gamma^2}\}$.

The above results show that in the presence of smart consumers, the main insights are similar to the insights in the basic model. The difference is that the degree of consumers' naïveté plays a role in the pricing decision and profit of the channel: both the equilibrium price and profit increases in γ. This is intuitive, when consumers are smarter, they are more likely to make accurate inferences about the true quality of the product in presence of review promotions, which makes the channel more difficult to improve the average rating. The interesting result is that, even if consumers are very smart (i.e., γ is small), the channel can always benefit from review promotions with an increased variance. However, when γ is very small, the promotion efficiency would decrease in the product quality. In this case, the channel selling a low-quality product gains more than the channel with a high-quality product, which echoes to the empirical findings of Hu et al. (2011). Differently, when the variance is reduced, the channel may be hurt

from review promotions, and the smarter the consumers are, the lower the probability of the channel to be better off with review promotions. Moreover, the third result even demonstrates that when consumers are extremely smart, the promotion strategy cannot help the channel to gain a higher profit, regardless of the quality of the product. Therefore, if the target consumers are quite smart or sophisticated, it is safer for the channel to increase the variance in the process of review promotions.

9. Conclusions and future research

In this paper, the impact of promotions of online reviews in a manufacturer–retailer channel has been examined using an analytical modeling framework. Two channel structures, centralized and decentralized, have been analyzed. The baseline assumption is that the retailer can take some reasonable strategies to promote or stimulate the average rating of online reviews in a reasonable range, such as providing some frills or coupons with some costs. In the process of such a promotion, the retailer can choose to increase or decrease the variance. Taking the situation without review promotions as the benchmark, the impact of promotions of online reviews on the equilibrium prices, as well as the demand and profitability of the retailer and manufacturer under centralized and the decentralized structures have been derived.

9.1. Main conclusions

We find that under both channel structures, the optimal degree of promotion increases with the quality of the product but decreases with the variance. If the retailer chooses to reduce the variance, the intuitive result is that a reduced variance and a higher average rating means the opinions of consumers about the value of the product are more consistent, which can increase their willingness to pay, thus having a positive impact on the demand. However, the results show that a reduced variance may hurt the profit of the channel and a channel with a lower-quality product has a higher probability of being worse off.

In contrast, if the variance is increased, although the optimal degree of promotion may be limited, the reduced price sensitivity drives the retailer to charge a higher retail price, which makes the channel gain more profit, regardless of the channel structure. However, if the promotion cost coefficient is relatively high, the promotion efficiency may decrease with the quality of the product. In other words, if the promotion cost is relatively high, then a channel with a higher-quality product gains less from the promotion strategy than does a channel with a lower-quality product.

Moreover, we have demonstrated that under the decentralized structure, the manufacturer has a greater threshold interval to benefit from promotions of online review than the retailer, which means that it is always wise for the manufacturer to take part in the promotion strategy. We also show that the channel efficiency can be improved by an increased variance, but a reduced variance may not always play a positive role in improving the channel efficiency of the decentralized channel.

Our results also have some implications for a more general setting of the market with traditional consumers who are not influence by online reviews. Most of results are still hold. One different result is that, if the channel has a relatively large size of traditional consumers, an increased variance may hurt the channel profit if the cost of the promotion strategy is relatively high.

9.2. Managerial implications

These results give insightful implications for practice. First, contrary to common wisdom, in the promotion of the average rating, it is better for the retailer to target those consumers who give favorable reviews (those whose ratings are higher than the average rating). Although this may affect the demand of the product negatively, the channel can always gain more profit with review promotions. Second, in practice, the retailer may be more likely to respond to those consumers who give extremely unfavorable reviews because having fewer unfavorable reviews can positively affect later consumers' evaluation of the product, increasing the demand for the product. Prior study reveals that some mining techniques are helpful to identify the influential reviewers (Li et al., 2010) who are used to give extreme reviews. In addition, the retailer can take some efforts to encourage consumer to give a higher rating in their follow-up reviews. However, our results demonstrate that the channel should be more careful to target these consumers. Third, under the decentralized channel, we suggest that the players can treat the promotion strategy as an effective cooperation contract to improve the channel efficiency.

9.3. Limitations and future work

Several limitations of the current study suggest future work. First, this paper has focused on the impact of online reviews on an experience product, and assumed that the average rating can reflect the true average value to the consumers. In practice, the value of the product may be affected by other information, such as advertising or free samples. The joint impact of online reviews and advertising would be a promising direction. In addition, we only examine the impact of promotions when online reviews are in a steady state. It would be more practical to model the promotion strategy in the long run. Further, there are several other coordination contracts for the manufacturer and the retailer to undertake promotional strategies, such as wholesale-price and revenue-sharing contracts. It would be interesting to compare the promotion efficiency under different types of contracts.

CRediT authorship contribution statement

Wenjuan Yang: Conceptualization, Methodology, Writing - original draft. **Jiantong Zhang:** Validation, Supervision, Writing - review & editing. **Hong Yan:** Visualization, Supervision, Writing - review & editing.

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Appendix A

Proof of Lemmas 1 and 2. Lemma 1 presents the equilibrium results of the benchmark without review promotions. Recall that the profit function in the centralized channel is

$$\pi_c(p_c) = \frac{(q + a - p_c)p_c}{2a}. \tag{A.1}$$

By solving the first-order conditions of Eqs. (A.1) for p_c, we obtain the equilibrium price $p_c = \frac{q+a}{2}$. Substituting this into the demand function (Eqs. (1)) and the profit function (Eqs. (A.1)), we get the results in Lemma 1.

With the same method, we derive the results with promotions of online reviews. Recall that the profit function with promotions in the centralized channel is

$$\pi_c^M(p_c^M, \eta_c) = \frac{(q + \eta_c + b - p_c^M)p_c^M}{2b} - \frac{c\eta_c^2}{2}. \tag{A.2}$$

By solving the first-order condition of Eqs. (A.2) for p_c^M and η_c, we obtain the equilibrium price $p_c^M = \frac{2(q+b)bc}{4bc-1}$ and the promotion level of the average rating $\eta_c = \frac{q+b}{4bc-1}$. Here, to ensure that $0 < \eta_c < 1$ always holds and to keep the analysis simple, we assume that c is not too small, and add the condition $c > \frac{b+2}{4b}$. It is easy to verify that when $c > \frac{b+2}{4b}$, the Hessian matrix is always negative definite. Substituting them into the demand function (Eqs. (2)) and the profit function (Eqs. (A.2)), we obtain the results in Lemma 2. □

Proof of Proposition 1. Note that $X_1 = \frac{c}{4bc-1}$. From Lemmas 1 and 2, we have

$$\Delta p_c = p_c^M - p_c = (2bX_1 - 1/2)q + 2b^2X_1 - a/2 \tag{A.3}$$

Eq. (A.3) shows that Δp_c is always increasing in q since $2bX_1 - 1/2 > 0$. In fact, when $2b^2X_1 - a/2 \geq 0$, we have $\Delta p_c \geq 0$. When $2b^2X_1 - a/2 < 0$, by solving $\Delta p_c = 0$, we get the threshold value $q_{c1} = \frac{a-4b^2X_1}{4bX_1-1}$; when $q < \min\{q_{c1}, 1\}$, we have $\Delta p_c < 0$. Combining the threshold values of the parameters before, we get the first result of this proposition.

Next, we turn to the comparison of the demand with and without promotions of online reviews. It is easy to get

$$\Delta D_c = D_c^M - D_c = (X_1 - 1/(4a))q + bX_1 - 1/4 \tag{A.4}$$

From Eq. (A.4), it is easy to check that $bX_1 - 1/4 > 0$. Thus, if $X_1 > 1/(4a)$, then $\Delta D_c > 0$. If $X_1 < 1/(4a)$, by solving $\Delta D_c > 0$, we have $q < \frac{a(4bX_1-1)}{1-4aX_1}$. Therefore, Proposition 1 holds. □

Proof of Proposition 2. From Lemmas 1 and 2, we have

$$\Delta\pi_c = \pi_c^M - \pi_c = \left(\frac{X_1}{2} - \frac{1}{8a}\right)q^2 + \left(bX_1 - \frac{1}{4}\right)q + \frac{b^2X_1}{2} - \frac{a}{8} \tag{A.5}$$

Let $A_1 = \frac{X_1}{2} - \frac{1}{8a}, B_1 = bX_1 - \frac{1}{4}, C_1 = \frac{b^2X_1}{2} - \frac{a}{8}$. We first treat the case when $b > a$. It is easy to check that $\frac{\partial^2\Delta\pi_c}{\partial q^2} < 0$, which means that $\Delta\pi_c$ is concave in q. In addition, if $q = 0$ or $q = 1$, then $\Delta\pi_c > 0$. Therefore, when $b > a$, the profit is always higher with the promotion. Further, by solving $\frac{\partial^2\Delta\pi_c}{\partial q^2} < 0$ and $\frac{\partial\Delta\pi_c}{\partial q} < 0$ at the same time, we get the result (1) of Proposition 2.

Next, we treat the case when $b < a$. In this case, $\frac{\partial\Delta\pi_c}{\partial q} > 0$. By solving $\Delta\pi_c = 0$, we get the threshold value $q = q_c$, where $q_c = \frac{a-2b\sqrt{aX_1}}{2\sqrt{aX_1}-1}$, and it is easy to check that when $c > \frac{a+1}{4(b-a)}$, it follows that $q_c > 0$.

Proof of Lemmas 3 and 4. Lemma 3 presents the equilibrium results of the benchmark under the decentralized channel. We obtain the equilibrium by backward induction. Recall that the demand function without the promotions is $D_d = \frac{q+a-p_d}{2a}$. The retailer's decision problem is as follows:

$$\max_{p_d} \ \pi_r = (p_d - w)\frac{q + a - p_d}{2a} \tag{A.6}$$

By solving the first condition of Eq. (A.6) for p_d, we obtain the response function for the retailer:

$$p_d^* = \frac{q + a + w}{2} \tag{A.7}$$

Substituting Eq. (A.7) into the demand function, we can then rewrite the decision problem of the manufacturer as

$$\max_{w} \ \pi_m = wD_d = w\frac{q + a - w}{4a} \tag{A.8}$$

Solving the first condition of Eq. (A.8) for w, we get $w^* = \frac{q+a}{2}$. Then, substituting this into Eq. (A.7) and the demand function, we obtain the outcomes in Lemma 3.

With the same method, we derive the results with promotions of online reviews under the decentralized channel. Note that in the presence of promotions, $D_d^M = \frac{q+\eta+b-p_d^M}{2b}$. We also rewrite the retailer's problem as

$$\max_{p_d^M,\eta_d} \pi_r^M = (p_d^M - w^M)\frac{q+\eta+b-p_d^M}{2b} - \frac{c}{2}(1-\lambda)\eta_d^2 \tag{A.9}$$

Then, we have $\frac{\partial^2\pi_r^M}{\partial {p_d^M}^2} = -\frac{1}{b} < 0$, $\frac{\partial^2\pi_r^M}{\partial {\eta_d}^2} = -c(1-\lambda) < 0$, and $\frac{\partial^2\pi_r^M}{\partial {p_d^M}^2}\frac{\partial^2\pi_r^M}{\partial {\eta_d}^2} - (\frac{\partial^2\pi_r^M}{\partial p_d \partial \eta_d})^2 = \frac{4bc(1-\lambda)-1}{4b^2}$. Note that π_r^M is jointly concave in p_d and η if and only if $4bc(1-\lambda)-1 > 0$. In order to avoid trivial problems and to focus on the impact of promotions, we only derive the results for this case and impose the above constraint throughout this paper. Then, calculating the first-order partial derivatives of Eq. (A.9) with respect to p_d^M and η_d, and setting the resulting expressions to zero, we obtain the following results:

$$\eta_d^* = \frac{w^M - q - b}{4bc\lambda - 4bc + 1}, \quad p_d^{M*} = \frac{(2bc\lambda - 2bc + 1)w^M + 2bc(\lambda-1)(b+q)}{4bc\lambda - 4bc + 1} \tag{A.10}$$

Substituting these into the demand function, and substituting the corresponding demand into the manufacturer's profit function, we can rewrite the decision problem of the manufacturer as follows:

$$\max_{w^M,\lambda} \pi_m^M = \frac{(1-\lambda)(w^M - q - b)cw^M}{4bc\lambda - 4bc + 1} - \frac{(w^M - q - b)^2 c\lambda}{2(4bc\lambda - 4bc + 1)^2} \tag{A.11}$$

Calculating the first-order partial derivatives of Eq. (A.11) with respect to w^M and λ, and setting the resulting expressions to zero, we obtain the following results:

$$\lambda^* = \frac{1}{3}, \quad w^{M*} = \frac{(q+b)(16bc-3)}{32bc-9}. \tag{A.12}$$

Then, substituting Eq. (A.12) into Eqs. (A.11), (A.10) and (A.9), we easily obtain the outcomes in Lemma 4. □

Proof of Proposition 3 and Proof of Proposition 4. From Lemmas 3 and 4, we have $\Delta w = w^M - w = (Y_1 - 1/2)q + bY_1 - a/2$. Note that Δw is always increasing in q since $Y_1 - 1/2 > 0$. By solving $\Delta w = 0$, we get the threshold value $q = \frac{a-2bY_1}{2Y_1-1}$. It is easy to check that when $b > a$, it follows that $\frac{a-2bY_1}{2Y_1-1} < 0$. When $b < a$ and $c > \frac{3(3a-2b)}{32b(a-b)}$, we have $\frac{a-2bY_1}{2Y_1-1} > 0$. Then we get result (1) of Proposition 3.

Next, we compare the profits of the manufacturer with and without review promotion. We have

$$\Delta\pi_m = \pi_m^M - \pi_m = \left(Y_3 - \frac{1}{16a}\right)q^2 + \left(2bY_3 - \frac{1}{8}\right)q + b^2Y_3 - \frac{a}{16} \tag{A.13}$$

Let $A_2 = Y_3 - \frac{1}{16a}$, $B_2 = 2bY_3 - \frac{1}{8}$, and $C_2 = b^2Y_3 - \frac{a}{16}$. We first check the condition when $b > a$. It is easy to check that $\frac{\partial^2\Delta\pi_m}{\partial q^2} < 0$, which means that $\Delta\pi_m$ is concave in q. In addition, when $q = 0$ or $q = 1$, we have $\Delta\pi_m > 0$. Therefore, when $b > a$, the profit is always higher with the promotion. Further, by solving $\frac{\partial^2\Delta\pi_m}{\partial q^2} < 0$ and $\frac{\partial\Delta\pi_m}{\partial q} < 0$ at the same time, we get the result (1) of Proposition 3.

Next, we treat the case when $b < a$. In this case, $\frac{\partial\Delta\pi_m}{\partial q} > 0$. By solving $\Delta\pi_m = 0$, we get the threshold value $q = q_m$, where $q_m = \frac{a-4b\sqrt{aY3}}{4\sqrt{aY3}-1}$, and it is easy to check that when $c > \frac{9a}{32b(a-b)}$, we have $q_m > 0$. Thus, the results of Proposition 3 hold.

Proposition 4 can be proved with the same logic, thus we omit the details here. □

Proof of Corollary 1. From Propositions 3 and 4, we find that when $b < a$, both the manufacturer and the retailer will be worse off. For this reason, we will treat the case when $b < a$. It is easy to check that $\frac{9(a+1)}{32(b-a)} < \frac{9(a+1)^2}{32(a-b)(ab-1)}$. In addition, when $\frac{9(a+1)}{32(b-a)} < c < \frac{9(a+1)^2}{32(a-b)(ab-1)}$, we have $0 < q_m < 1$ and $q_m < q_r$. Thus, Corollary 1 holds. □

Proof of Proposition 5. From Lemmas 1 to 4, we have $\Delta D_c = \frac{(1+4ac-4bc)q}{4(4bc-1)a} + \frac{1}{4(4bc-1)}$ and $\Delta D_d = \frac{(9+32ac-32bc)q}{8a(32bc-9)} + \frac{9}{8(32bc-9)}$. Proposition 5 can be easily derived by solving $\Delta D_c < 0$ and $\Delta D_d < 0$. □

Proof of Proposition 6. We use Ef and Ef^M to denote the channel efficiency of the decentralized channel. Then from Lemmas 1 to 4, we have $Ef = \frac{\pi_m+\pi_r}{\pi_c}$ and $Ef^M = \frac{\pi_m^M+\pi_r^M}{\pi_c}$. Let $\Delta_{Ef} = Ef^M - Ef$. With the same method as in the proof of Proposition 2, we can get the result of Proposition 6. □

Appendix B

Proof of Proposition 7. It is easy to derive the profit function of considering traditional consumers:

$$\pi_c(p_c) = (1-k)\frac{(q+a-p_c)p_c}{2a} + k(1-p_c). \tag{B.1}$$

By solving the first-order conditions of Eqs. (B.1) for p_c, we obtain the equilibrium price $p_c = \frac{ak+a-kq+q}{2(2ak-k+1)}$. Substituting this into the profit function (Eqs. (B.1)), we get the equilibrium results as follows.

$$D_c = \frac{ak+a-kq+q}{4a}, \pi_c = \frac{(ak+a-kq+q)^2}{8((2a-1)ak+a)}. \tag{B.2}$$

With the same method, we derive the results with promotions of online reviews. The profit function with promotions in the centralized channel can be characterized as

$$\pi_c^M(p_c^M, \eta_c) = (1-k)\frac{(q+\eta_c+b-p_c^M)p_c^M}{2b} - \frac{c\eta_c^2}{2} + k(1-p_c^M). \tag{B.3}$$

By solving the first-order condition of Eqs. (B.3) for p_c^M and η_c, we obtain the equilibrium price $p_c^M = \frac{2bc(bk+b-kq+q)}{8b^2ck+4bc(1-k)+(1-k)^2}$ and the promotion level of the average rating $\eta_c = \frac{(1-k)(bk+b-kq+q)}{8b^2ck+4bc(1-k)+(1-k)^2}$. Substituting them into the demand function (Eqs. (2)) and the profit function (Eqs. (B.3)), we obtain the optimal demand and profit with review promotions as follows.

$$D_c^M = \frac{c((2b-1)k+1)(bk+b-kq+q)}{8b^2ck+4bc(1-k)+(1-k)^2}, \pi_c^M = \frac{c(bk+b-kq+q)^2}{16b^2ck+8bc(1-k)-2(1-k)^2} \tag{B.4}$$

From the equilibrium results before, we first compare the optimal prices. Let $\Delta p_c = p_c^M - p_c = \frac{2bc(bk+b-kq+q)}{8b^2ck+4bc(1-k)+(1-k)^2} - \frac{ak+a-kq+q}{2(2ak-k+1)}$. It is easy to check that when $b > a$, $\Delta p_c > 0$ always holds. When $b < a$, we have Δp_c increases with q. By solving $\Delta p_c = 0$, we obtain the unique threshold value $q = \frac{(k+1)(a(4bc+k-1)-4b^2c)}{k(8bc(a-b)+k-2)+1}$, which is larger than zero only when $c > \frac{a-ak}{4ab-4b^2}$.

Next, we compare the optimal profits with and without review promotions. Let $\Delta\pi_c = \pi_c^M - \pi_c = \frac{c(bk+b-kq+q)^2}{16b^2ck+8bc(1-k)-2(1-k)^2} - \frac{(ak+a-kq+q)^2}{8((2a-1)ak+a)}$. When $b < a$, we find that $\Delta\pi_c$ increases with q. By solving $\Delta\pi_c = 0$, we get the unique threshold value $q = q_e = 2\sqrt{\frac{ac(k+1)^2(a-b)^2((2a-1)k+1)(8b^2ck-4bc(k-1)-(k-1)^2)}{(k-1)^2(4ck(a-b)(2a+2b-1)+4ac-4bc+k^2-2k+1)^2}} + \frac{a(k+1)(k(8bc(a-b)+k-2)+1)}{(k-1)(4ck(a-b)(2a+2b-1)+4ac-4bc+k^2-2k+1)}$, which is large than zero only when $k > \frac{a-ak}{4ab-4b^2}$. Differently, when $b > a$, we find that when $0 < k < \sqrt{\frac{4a^2b^2+a^2-2ab+b^2}{(3ab-2a-2b+1)^2}} + \frac{a(-b)-a-b+1}{3ab-2a-2b+1}$, $\Delta\pi_c > 0$ always holds. Otherwise, $\Delta\pi_c < 0$ if and only if $c > \frac{(1-k)((a-1)k+a+1)^2}{4(b-a)(k^2(a(3b-2)-2b+1)+2k(ab+a+b-1)-ab+1)}$ and $q > q_e$. □

Proof of Proposition 8. In the extension of considering smart consumers, we can obtain the demand of consumers $D^M = \frac{q+\gamma\eta_c+b-p_c^M}{2b}$. Then the profit function is as follows:

$$\pi_c(p_c^M) = \frac{(q+\gamma\eta+b-p_c^M)p_c^M}{2b} \tag{B.5}$$

By solving the first-order conditions of Eqs. (B.5) for p_c^M and η_c^M, we obtain the equilibrium results: $p_c^M = -\frac{2bc(b+q)}{\gamma^2-4bc}$ and $\eta_c = \frac{\gamma(b+q)}{4bc-\gamma^2}$. Substituting them into the profit function (Eqs. (B.5)), we get the equilibrium results as follows.

$$D_c^M = \frac{c(b+q)}{4bc-\gamma^2}, \pi_c^M = \frac{c(b+q)^2}{8bc-2\gamma^2}. \tag{B.6}$$

With the equilibrium results, we first compare the optimal prices. Let $\Delta p_c = p_c^M - p_c = \frac{2bc(b+q)}{4bc-\gamma^2} - \frac{a+q}{2}$. It is easy to check that when $b > a$, $\Delta p_c > 0$ always holds. When $b < a$, we have Δp_c increases with q. By solving $\Delta p_c = 0$, we obtain the unique threshold value $q = \frac{4abc-a\gamma^2-4b^2c}{\gamma^2}$, which is larger than zero only when $c > \frac{a\gamma^2}{4ab-4b^2}$.

Next, we compare the optimal profits with and without review promotions. Let $\Delta\pi_c = \pi_c^M - \pi_c = \frac{c(b+q)^2}{8bc-2\gamma^2} - \frac{(a+q)^2}{8a}$. When $b > a$, we find that $\Delta\pi_c > 0$ always holds. It is easy to check that when $c > \max\{\frac{(a+1)\gamma^2}{4(b-a)}, \frac{b+2}{4b}\}$ and $-\frac{a\gamma^2}{4ac-4bc+\gamma^2} < q \le 1$, $\frac{\partial\Delta\pi_c}{\partial q} < 0$. When $b < a$, we find that $\Delta\pi_c$ increases with q. By solving $\Delta\pi_c = 0$, we get the unique threshold value $q = q_s = \frac{2(a-b)\sqrt{ac(4bc-\gamma^2)}}{4ac-4bc+\gamma^2} - \frac{a\gamma^2}{4ac-4bc+\gamma^2}$, which is only large than zero when $c > \frac{a\gamma^2}{4ab-4b^2}$. Besides, we have $q_s > 1$ if $\gamma < \frac{1}{a+1}\sqrt{\frac{(b+2)(a-b)(ab-1)}{b}}$ or if $c > \frac{(a+1)^2\gamma^2}{4(a-b)(ab-1)}$. □

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